PROJECT OUTCOMES

Project Success and Technical Achievements

- Successfully achieved the main goal of making quantum computing concepts easier to understand through visual representation.
- Developed a working Python prototype that converts quantum circuits into ZX-style diagrams.
- Implemented support for key quantum gates, including Hadamard, Z, and CNOT.
- Created a system of coloured “spiders” that stores phase values, models spider fusion, and represents qubit transformations visually.
- Demonstrated that quantum logic can be expressed through diagrams rather than relying solely on mathematical equations.

Educational Value and Future Applications

- Showed that visual representations make abstract concepts such as superposition and probabilistic measurement outcomes easier to understand.
- Verified that the generated ZX-style diagrams accurately reflect the structure and behaviour of the original quantum circuits.
- Highlighted potential uses in schools, universities, STEM clubs, and outreach programmes as a teaching aid.
- Identified opportunities for future development, such as direct Qiskit integration, automatic circuit-to-diagram conversion, and interactive web-based visualisation tools.

Impact, Sustainability and Overall Evaluation

- Promotes accessible digital learning without requiring specialist hardware or physical resources.
- Supports environmentally sustainable, paperless education through software-based learning tools.
- Demonstrates that visual reasoning can simplify complex scientific topics and encourage engagement with quantum computing.
- Concludes that the project successfully fulfilled its original aim of making quantum logic more approachable, creative, and understandable for beginners through interactive visualisation.